CAPSTONE PROJECT

ATLANTIC CANADA CRIME PATTERNS ANALYSIS: CLUSTERING LOCALITIES BY TYPE, STRUCTURE, AND GROWTH (2020–2024)

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PROBLEM DEFINITION

Identify patterns of criminal activity and grouplocalities within Atlantic Canadian provinces based on similarities in crime trends between 2020 and 2024.

Provinces:

- Nova Scotia
- New Brunswick
- Newfoundland and Labrador
- Prince Edward Island

CONTEXT

- Publicly available crime data from Statistics Canada
  https://www150.statcan.gc.ca/t1/tbl1/en/cv.action?pid=3510017801

- Detailed counts of Criminal Code violations

- Data organized by localities and years (2020–2024)

STATISTICS CANADA

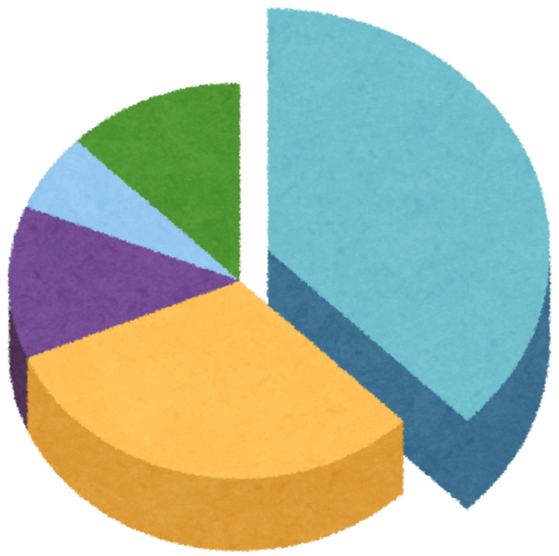
Format file:

- Total, all violations
  - Total, all Criminal Code violations (including traffic)
    - Total, all Criminal Code violations (excluding traffic)
      - Total violent Criminal Code violations
      - Total property crime violations
      - Total other Criminal Code violations
    - Total Criminal Code traffic violations
      - Total impaired driving
      - Total other Criminal Code traffic violations
  - Total Federal Statute violations
    - Total Drug violations
      - Possession, cannabis (pre-legalization)
      - Possession, cocaine
      - Total, possession, other Controlled Drugs and Substances Act drugs
      - Total cannabis, trafficking, production or distribution (pre-legalization)
      - Total cocaine, trafficking, production or distribution
      - Total other Controlled Drugs and Substances Act drugs, trafficking, production or distribution
      - Possession, sale, etc., for use in production of or trafficking in substance
    - Total Cannabis Act
    - Youth Criminal Justice Act
    - Total other Federal Statutes
      - Bankruptcy Act
      - Income Tax Act
      - Canada Shipping Act
      - Canada Health Act
      - Customs Act
      - Competition Act
      - Excise Act
      - Total Immigration and Refugee Protection Act
      - Firearms Act
      - National Defence Act
      - Quarantine Act
      - Other federal statutes

WHY IS THIS IMPORTANT?

Raw numbers do not always show patterns clearly.

- Two provinces may have different total crime levels.
- But they may share similar proportions of crime types.
- Or they may show similar growth trends over time.



WHO BENEFITS AND WHY?

- Help policymakers allocate resources more effectively.
- Support researchers in comparative studies.
- Improve public understanding of crime trends.

APPROACH & TOOL SELECTION

1. Data Processing and Exploration
2. Feature Engineering & Normalization
3. Unsupervised Machine Learning (Clustering)
4. Interpretation and Reporting

TOOLS, METHODS, AND TECHNIQUES

Programming Environment: Python

- **Pandas**: Data manipulation and cleaning
- **NumPy**: Numerical operations
- **Matplotlib**: Adjustments and custom plot details.
- **Seaborn**: Main visualizations for style and ease.
- **Scikit-learn**: Machine learning (K-means, PCA, scaling)

Data Processing Techniques

- Data cleaning and restructuring
- Aggregation
- Feature engineering (totals and proportions)
- Feature scaling

Machine Learning Technique

- K-means clustering
- Silhouette score for validation
- PCA for visualization
- Generative AI Usage

VISUALIZING CRIME CLUSTERS

Showing clusters of cities by total crime, crime distribution, and crime growth:

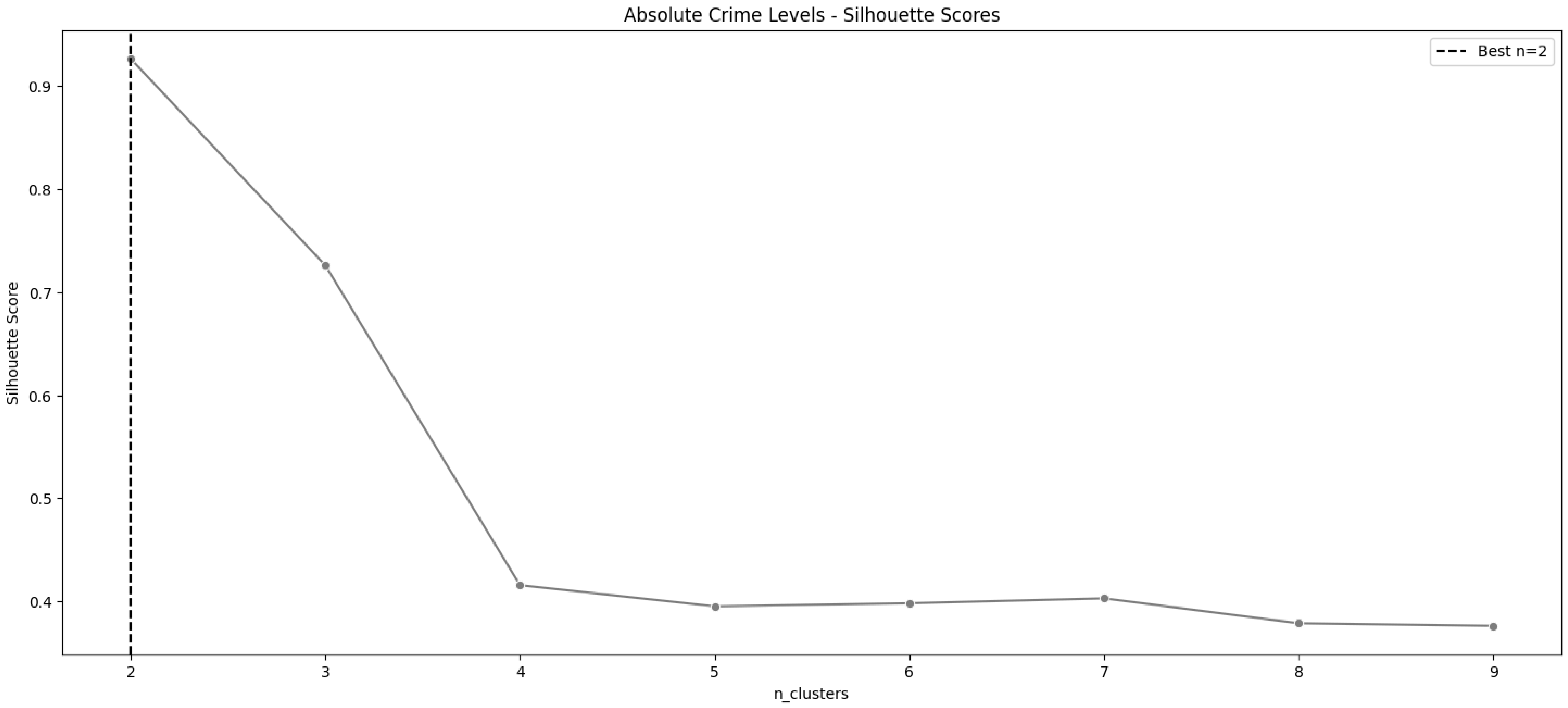
- **Absolute crime levels:**
  → Total number of crimes.
- **Proportional structure:**
  → Percentage of each crime type within a locality.
- **Growth patterns:**
  → Year-to-year change in crime rates.

MACHINE LEARNING

- K-means clustering
- Silhouette score to evaluate cluster quality
- PCA to visualize the results in two dimensions

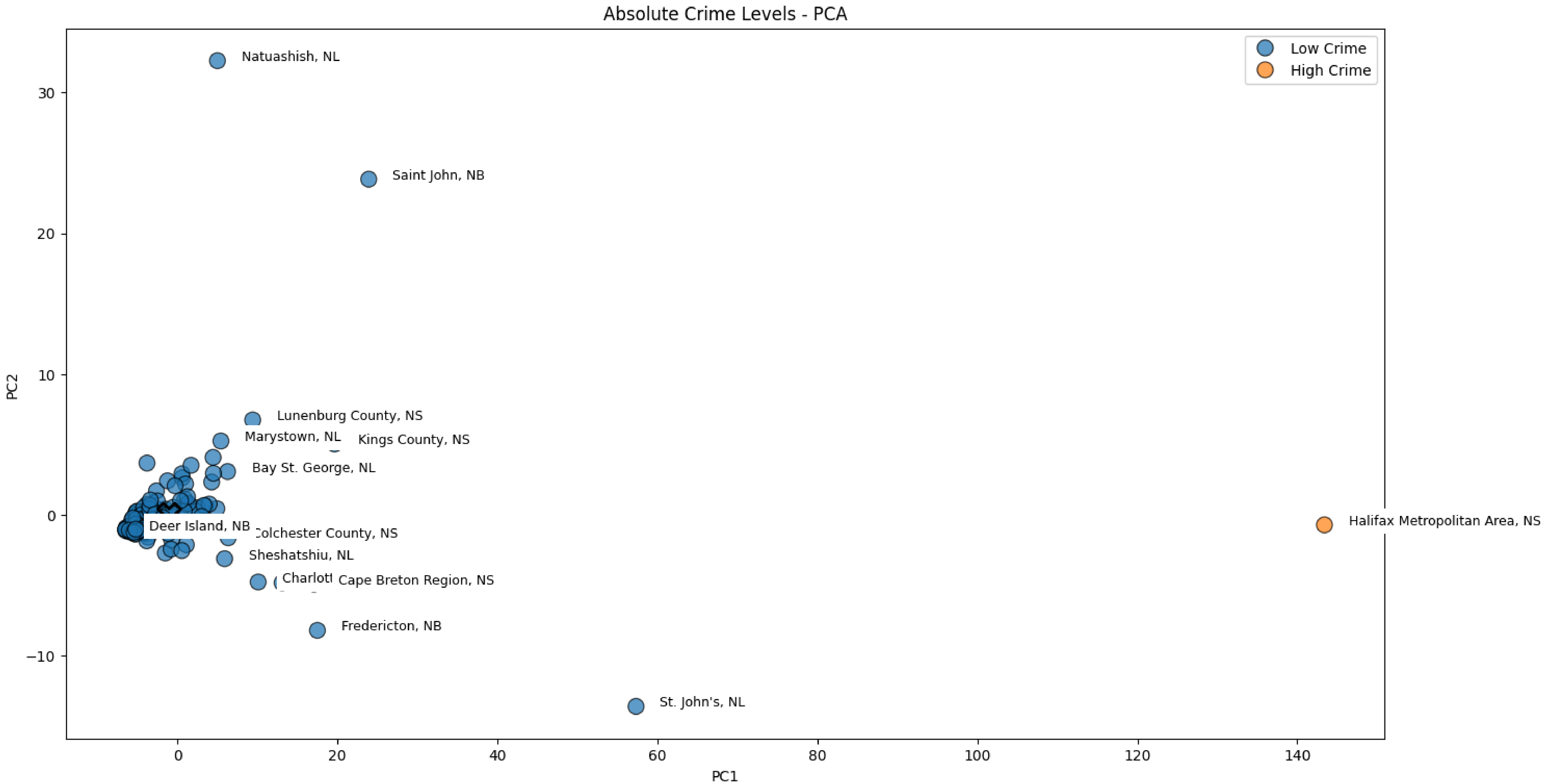
SILHOUETTE SCORES

Absolute Crime Levels



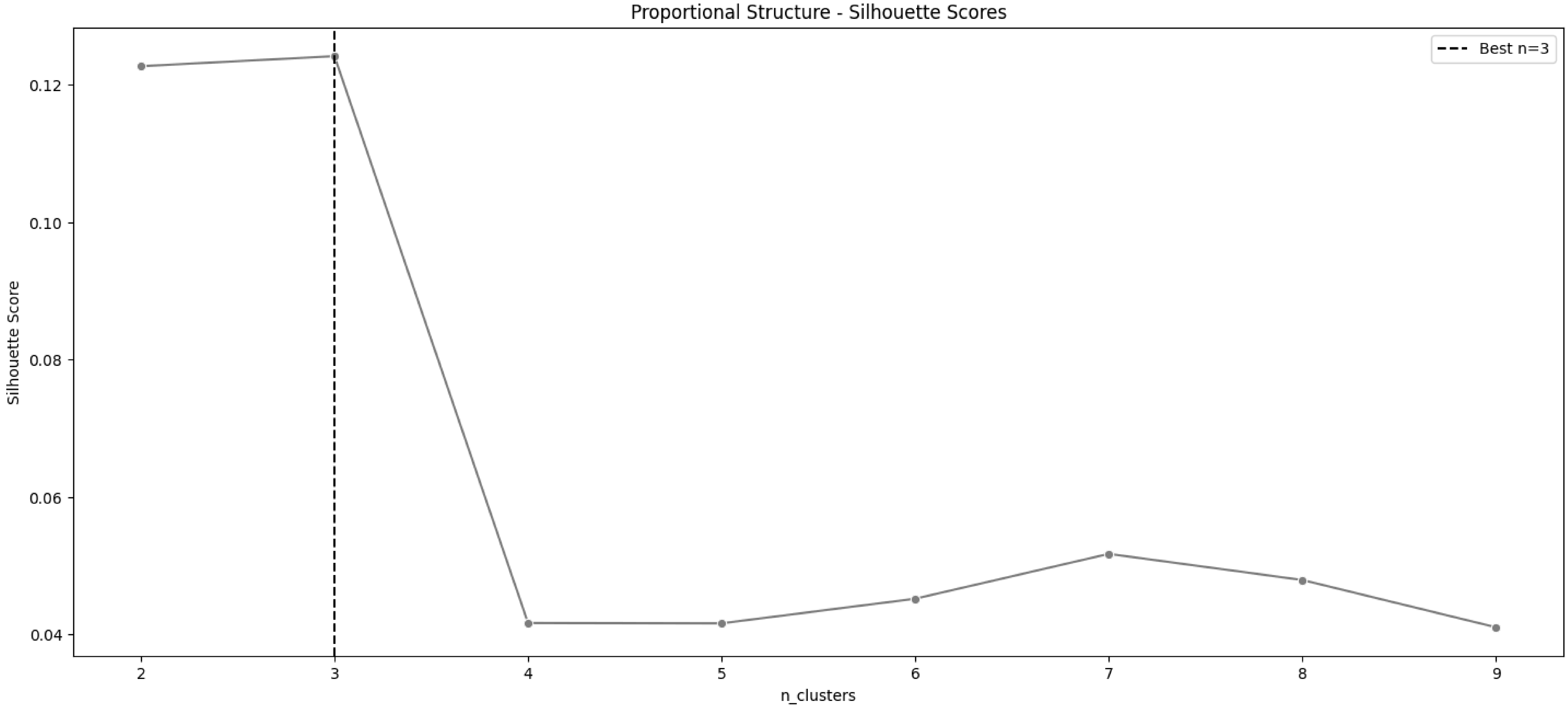
PCA SCATTER

Absolute Crime Levels



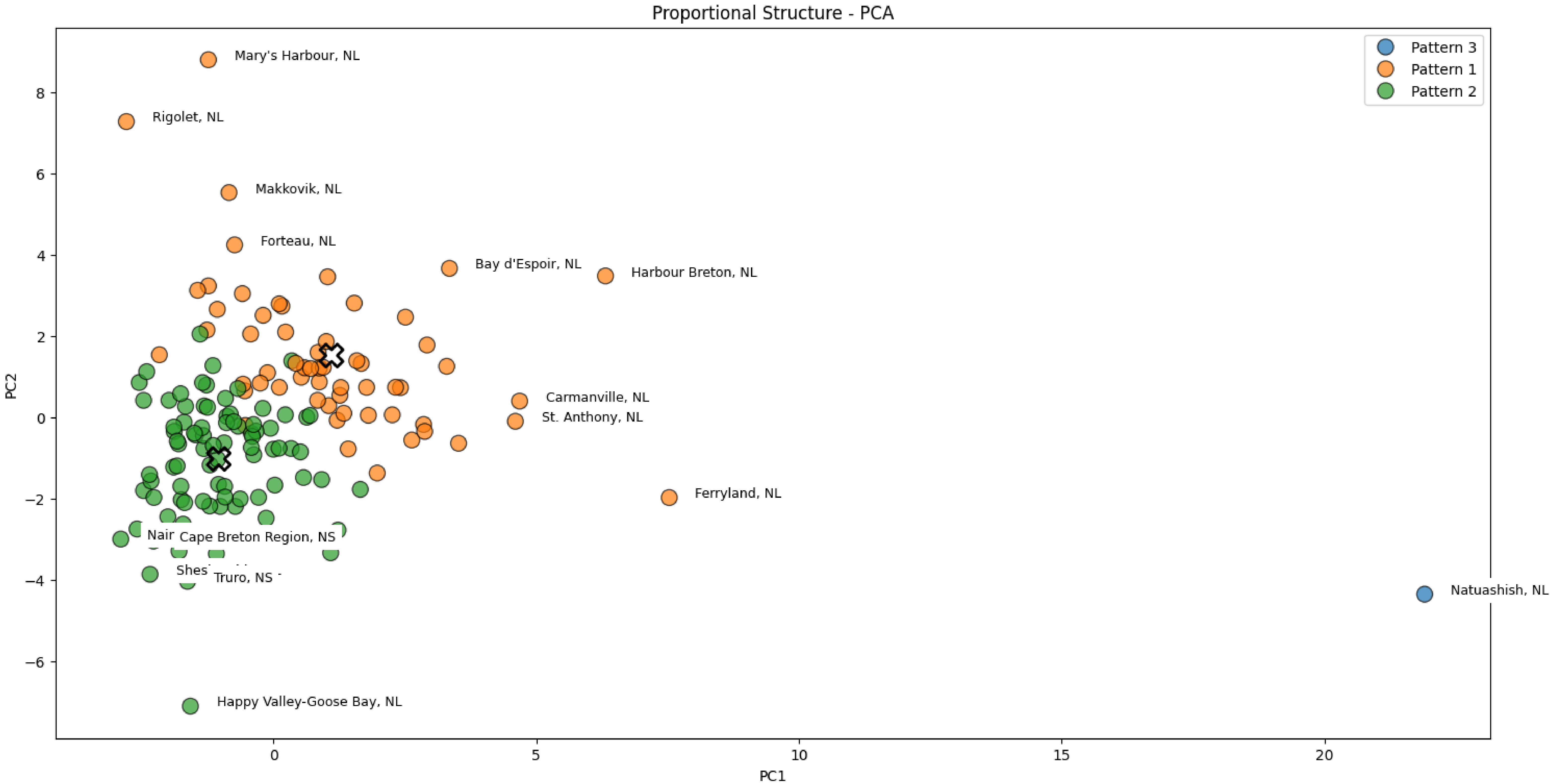
SILHOUETTE SCORES

Proportional Structure



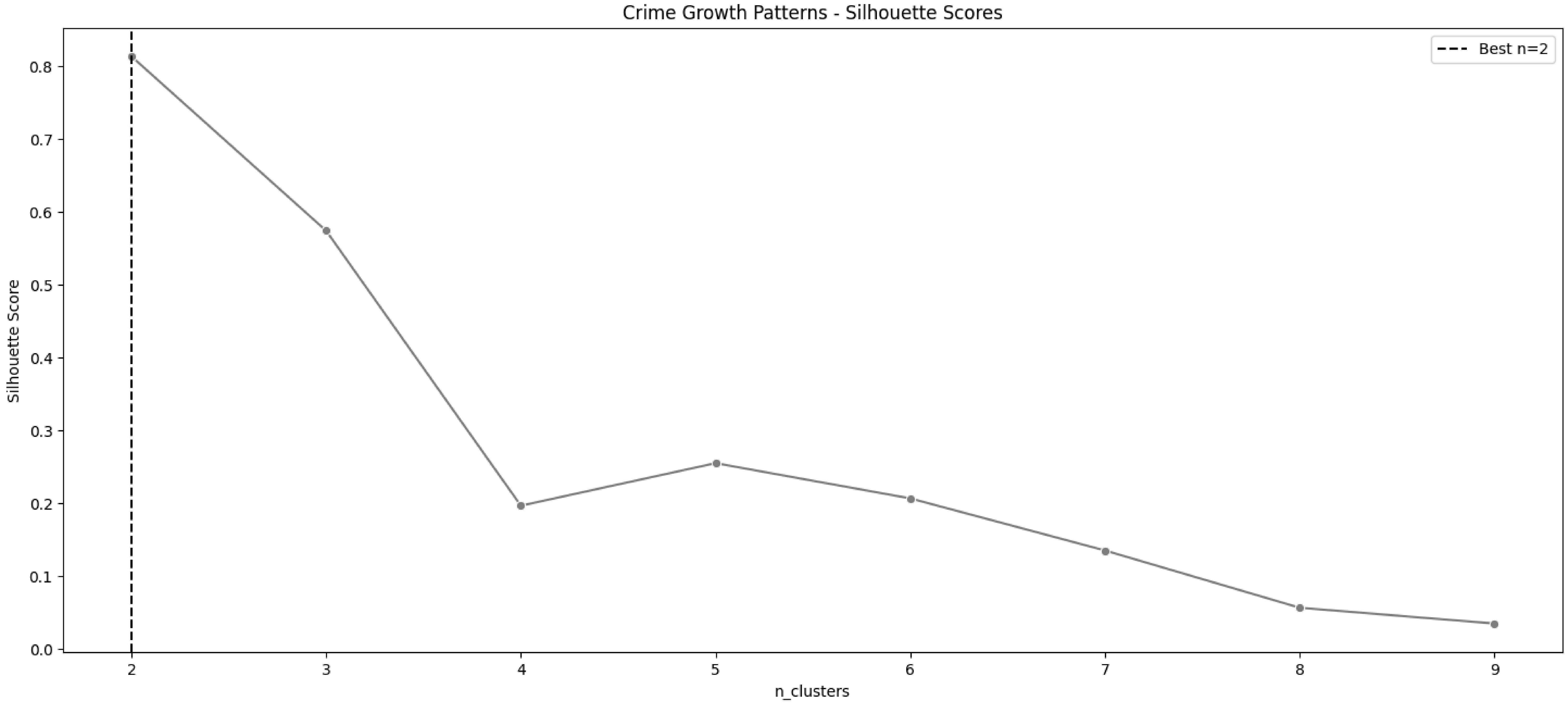
PCA SCATTER

Proportional Structure



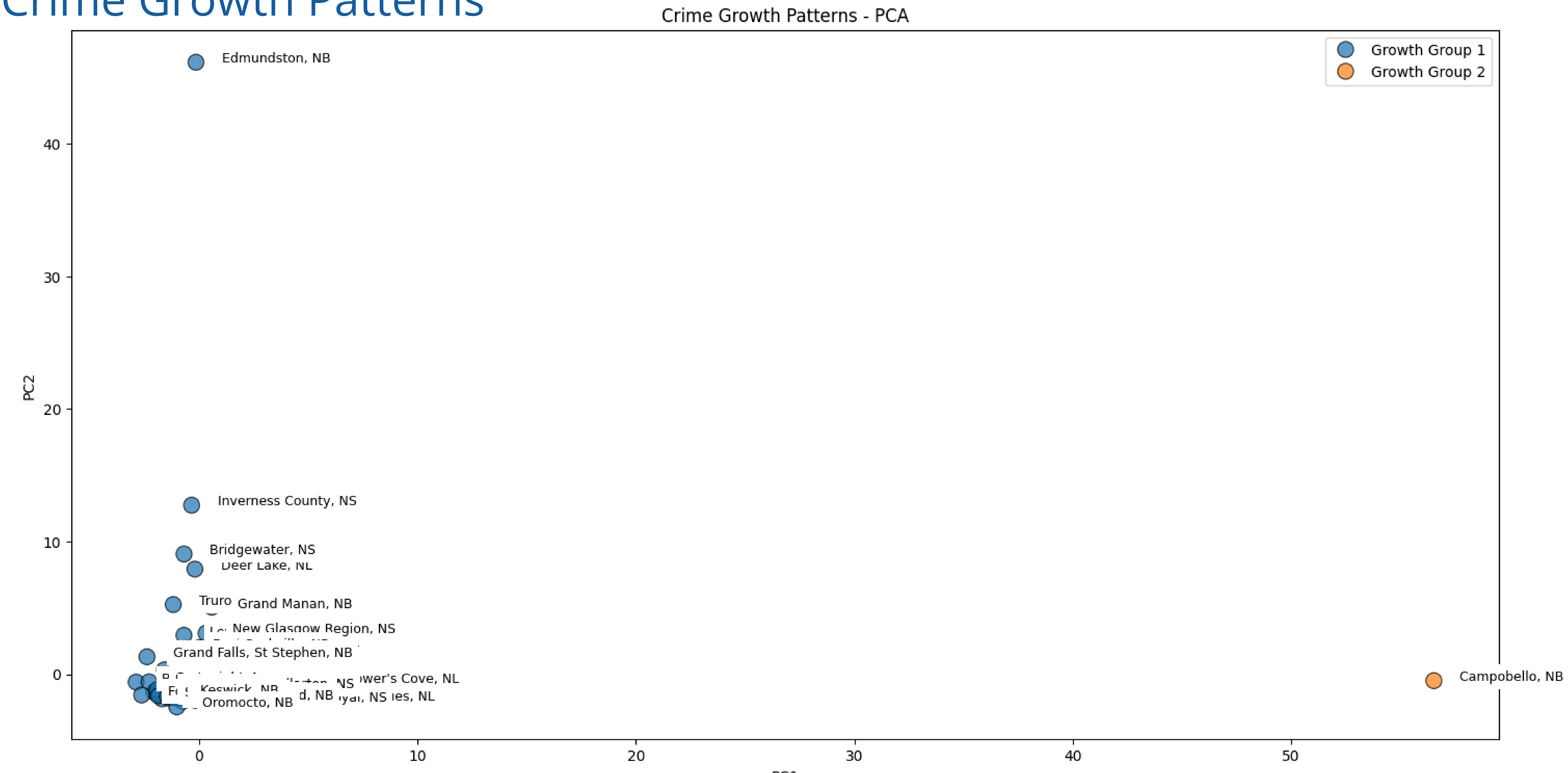
SILHOUETTE SCORES

Crime Growth Patterns



PCA SCATTER

Crime Growth Patterns



RESULTS OF THE CRIME ANALYSIS

- Most localities grouped into one large cluster, reflecting very similar crime structures.
- A smaller cluster included localities with distinct patterns
- Most Atlantic localities are structurally similar in terms of crime composition

CHALLENGES & LIMITATIONS

- Most localities are very similar, so clusters are not strongly separated.
- The dataset only includes crime counts — no demographic data.
- Only five years of data were available.
- Scaling and normalization were essential to avoid biased results.

CONCLUSIONS

- Data preparation is critical
- Proportions can be more meaningful than totals
- Machine learning does not replace human interpretation
- Generative AI helped structure code and documentation